

Umair Hassan

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Education	Punjab University College of IT (PUCIT) , Lahore, Pakistan Bachelor of Science in Software Engineering CGPA: 3.71 / 4.00 Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Databases, Machine Learning, Natural Language Processing	2020–2024
Publications	Umair Hassan. <i>COCO-Urdu: A Large-Scale Urdu Image-Caption Dataset with Multimodal Quality Estimation</i> . Submitted to ICLR 2026. arXiv:2509.09014 .	
Industry Experience	Noon , SDE-II Remote, Dubai, UAE - Designed and built a task-assignment engine in Python that dynamically allocates warehouse shipments based on shipment priority, volume, status, worker availability, experience, and upcoming breaks; reduced worker idle time by 13% and shipment processing time by 9% across 2 Dubai hubs. - Integrated the assignment engine across workforce and attendance microservices using GCP Pub/Sub, Spanner, and REST APIs, developing background workers for asynchronous data synchronization and expanding automated workforce coverage to 85%. - Built Python/FastAPI services providing real-time shipment volume, tracking, worker availability, and status across complex shipment workflows; optimized MySQL queries and implemented caching to deliver millisecond-level responses while processing thousands of shipments. - Developed reusable skills and plugins for AI-powered development tools, enabling engineers to automate repetitive workflows and improve developer productivity across backend development tasks.	06/2026–Present
	Motive Inc. (formerly KeepTruckin) , Backend Software Engineer (L3) Remote, San Francisco, CA - Led the design and development of Smart Notifications, a real-time decision system combining GPS telemetry, fuel data, geometric reasoning, and Generative AI models to optimize fleet fuel spending. - Built secure SFTP/FTP ingestion pipelines (1M records/day) with AWS S3, SQS, Docker, and Kubernetes, enabling scalable data feeds for ML and analytics services. - Integrated an OCR model into Mastercard transaction spend breakdown using customer receipts, boosting categorization accuracy and reducing financial discrepancies. - Developed and optimized Motive’s AI-powered fraud detection system achieving 90% accuracy, enabling the company’s \$250K/year Fraud Protection Guarantee by ensuring high model precision, low latency, and robust data integrity. - Served as primary on-call engineer for transaction gateways, pricing pipelines, and APIs, leveraging Datadog, Kibana, and Python to triage and resolve live issues under strict SLAs.	10/2024–06/2026
Projects	Kamaee GitHub - Built and shipped a mobile-first, WhatsApp-first business management platform for small shops in Pakistan, replacing traditional POS workflows with offline inventory, barcode scanning, customer credit ledgers, cash reconciliation, returns, receipts, and profit analytics. - Designed an employee-theft detection system for unrecorded sales by comparing physical inventory counts against transaction-derived stock levels, detecting discrepancies that conventional cash reconciliation cannot identify without requiring cameras, hardware, or employee authentication.	2026

- Designed a WhatsApp AI assistant that lets shopkeepers query their business in plain Urdu, including voice notes, replacing traditional dashboards with a familiar conversational interface for users with limited digital literacy.
- Implemented AI safety through application architecture, constraining the model to structured tool selection and typed refusal outputs so it cannot generate unrestricted responses or answer outside the supported business workflows.

Valorant Object Detection with Google PaliGemma 2025
[GitHub](#) [Hugging Face](#)

- Fine-tuned Google PaliGemma-2 (3B multimodal model) using QLoRA (4-bit quantization) for domain-specific object detection in Valorant gameplay.
- Built inference pipeline with Hugging Face Processor + Model APIs; evaluated performance using IoU, mAP@[0.50:0.95], AP@0.50/0.75, Accuracy, and Recall.
- Achieved 61% improvement in overall mAP; largest gains on small objects (+75%) indicating strong detection of critical small targets.
- Released fine-tuned model publicly on Hugging Face for reproducibility and benchmarking.

ViT-Insight: Vision Transformer Attention Explorer 2025
[GitHub](#)

- Developed an interactive platform to visualize per-layer attention maps and compute Attention Rollout in ViT models, tracing how input patches contribute to predictions across layers and heads.
- Enabled multi-label support, layer-range selection, and customizable heatmaps for detailed interpretability of individual images and classes.
- Generated dynamic GIFs demonstrating layer-wise attention evolution, enhancing transparency and trust in transformer-based vision models.

Other Experience	CC-211 Object Oriented Programming, <i>Teaching Assistant</i>	Spring 2024
	CC-211-L Object Oriented Programming Lab, <i>Teaching Assistant</i>	Spring 2024

Achievements

- Represented Pakistan at the **ICPC Asia West finals** – Honorable Mention [\[Certificate\]](#).
- Secured **Gold (2024) and Silver (2025) medals** at ICPC Asia Topi Regionals [\[Gold\]](#) [\[Silver\]](#).
- Ranked 7th nationally (solo) in Meta HackerCup 2023.
- Recognized among the Top 1% IT graduates by HEC Pakistan.
- Multiple podium finishes in national programming competitions (PUCON'23, INMIC'24, DAIRA'24, Electrocon'24, Technocon'24, BrainX'24) [\[LinkedIn Awards\]](#).

Community Involvement	Punjab Pakistan Flood Relief, <i>Volunteer (Humanitarian Response)</i> , Alkhidmat Pakistan	2025
	Türkiye Earthquake Relief Fundraising, <i>Volunteer</i> , Alkhidmat Pakistan	2023
	PUCON'24 Programming Competition, <i>Judge & Problem Setter</i>	2024
	CodeBees'24 Programming Competition (Women-only), <i>Judge & Problem Setter</i>	2024
	Google Developers Student Club PUCIT, <i>Mentor</i>	04/2024–06/2024

Technical Skills

AI/ML: PyTorch, scikit-learn, OpenCV, model training & evaluation
Programming: Python, C++, Ruby, Go
Systems/Cloud: AWS (S3, SQS, EC2, RDS), Datadog, PostgreSQL, Redis, Docker
Strengths: Algorithmic problem-solving, research-driven experimentation, computer vision, multi-modal systems

References

Dr. Syed Muhammad Ali
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Prof. Abdul Mateen
 Assistant Professor of Computer Science at PUCIT Lahore, **Email:** amateen@pucit.edu.pk